

Ryan Cho

Senior Software Engineer | Full-Stack Python & Distributed Systems

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PROFESSIONAL SUMMARY

Software engineer with 8+ years building production full-stack applications across finance and technology. Deep expertise in Python APIs, relational data modeling, and containerized deployments on Kubernetes. Proven track record designing event-driven services, ETL workflows, and performant web interfaces with TypeScript. Collaborative technical leader who partners with product and data teams to ship scalable SaaS platforms.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and maintained **Python microservices** powering portfolio analytics and reporting for internal stakeholders, reducing manual reconciliation cycles by 40% through **automated validation pipelines**.
- Built **RESTful APIs** with SQLAlchemy ORM layers over PostgreSQL, enforcing schema migrations and query optimization that cut average response latency from 800ms to 120ms.
- Developed **TypeScript and Vue** dashboard modules for real-time holdings visualization, integrating WebSocket feeds and improving workflow efficiency across three business units.
- Designed **event-driven ingestion services** using message queues to process market data streams, achieving 99.9% uptime for downstream risk calculation jobs.
- Implemented **ETL pipelines** transforming third-party financial feeds into warehouse-ready datasets, processing 2M+ records daily with idempotent retry logic and audit trails.
- Containerized applications with **Docker** and deployed via **Helm charts to Kubernetes** clusters across dev, staging, and production, standardizing release workflows for the engineering team.
- Partnered with **data scientists and product managers** to define API contracts and feature roadmaps, translating complex stakeholder requirements into iterative full-stack deliverables.
- Established **CI/CD guardrails** and infrastructure-as-code practices with Terraform modules, reducing environment provisioning time and improving deployment consistency.
- Mentored two junior engineers on **Python best practices**, code review standards, and production debugging techniques, elevating team delivery velocity.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Delivered **Python automation tools** for trading operations teams, replacing manual spreadsheet workflows and saving 15+ analyst hours per week across fixed-income desks.
- Engineered **SQL-driven reporting pipelines** against enterprise relational databases, supporting regulatory submissions with auditable data lineage and version-controlled query libraries.
- Built internal **web applications** connecting front-office systems to middle-office validation layers, improving cross-desk data consistency for thousands of daily transactions.
- Collaborated with infrastructure teams to deploy services on **containerized Linux environments**, gaining production experience with Docker-based release patterns.
- Developed **asynchronous batch jobs** for end-of-day reconciliation, processing position and P&L datasets with fault-tolerant scheduling and alerting integrations.
- Participated in **agile sprint ceremonies** and cross-functional design reviews, communicating technical tradeoffs to business stakeholders in clear, actionable terms.
- Contributed to **API integration layers** linking legacy mainframe exports to modern Python services, enabling incremental modernization without disrupting trading windows.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Implemented **REST API endpoints** in Python supporting inventory management features, enabling real-time stock updates for restaurant clients across 50+ locations.
- Built **React UI components** for admin dashboards, improving data entry workflows and reducing form completion errors through client-side validation.
- Wrote **unit and integration tests** covering critical payment and order flows, increasing release confidence for weekly deployment cadence.
- Assisted senior engineers in **PostgreSQL schema design** and migration scripts, learning production data modeling patterns for multi-tenant SaaS products.
- Participated in code reviews and **Git-based branching workflows**, contributing documentation and bug fixes that accelerated team onboarding.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive web pages** for nonprofit fundraising campaigns using modern HTML, CSS, and JavaScript, improving mobile accessibility for donor flows.
- Integrated **third-party donation APIs** into the foundation website, enabling secure payment processing and real-time contribution tracking.
- Collaborated with designers to translate wireframes into **production-ready layouts**, maintaining cross-browser compatibility and WCAG-aligned markup.
- Configured **content management updates** and deployment scripts, supporting rapid iteration on campaign landing pages during peak fundraising periods.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: TypeScript, Vue.js, React, JavaScript, HTML5, CSS3, Responsive Web Design

Backend: Python, FastAPI, asyncio, REST APIs, SQLAlchemy, Event-Driven Architecture

Databases: PostgreSQL, SQL, Snowflake, Relational Data Modeling, Query Optimization

Cloud & DevOps: Docker, Kubernetes, Helm, Terraform, Linux, CI/CD Pipelines

Monitoring & Tools: Grafana, Prometheus, Git, pytest, WebSockets, Logging Frameworks

Workflow & PM: Argo Workflows, Agile/Scrum, Jira, Technical Documentation, Cross-functional Collaboration